INTERMEDIATE PROJECT REPORT

TERMS OF REFERENCE FOR AN URBAN INTERVENTION

PROJECT WORK #2

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0 ABSTRACT

Our cities are changing constantly, and so our way to read the spaces that surround it should adapt to these changes. With this in mind, this project aims to develop a comprehensive strategy of intervention for a specific consolidated area in Porto, taking into consideration the concepts of legibility, permeability, variety and robustness.

The analysis of the space is enabled by the understanding of the issues detected in the aforementioned aspects, and also with the acknowledgement of the potential opportunities in the insertion in the urban tissue.

1 AIM

The aim of the proposed project is to expand the practical exercise of reading a city through analysis, discussions with different backgrounds and a diagnosis based on the assumption of improving the space for the city users.

The purpose of this report is to deliver an update on the progress made on the aforementioned project and elaborate a preliminary program, based on the structural insertion of the area in a bigger scale, and its access and memory of the place.

2 INTERVENTION AND STUDY AREAS

INTERVENTION AREA CHARACTERIZATION

UOPG, which stands for *Unidades Operativas de Planeamento e Gestão* (Operative Units of Planning and Management) correspond to the territorial polygons established in the PDM, *Plano Director Municipal* (Municipal Director Plan). They are endowed with programmatic content that guides and promotes its implementation.

The intervention area chosen for this project is UOPG 12, *Doze Casas* (Figure 1). This unit comprises a block delimited by *Rua de Santa Catarina*, *Rua da Alegria*, *Rua das Doze Casas* and *Rua da Escola Normal*. It is entirely located in the parish of Bonfim.

According to the PDM, the area must be occupied mainly by public and private green spaces and equipments, and it can be admitted the punctual construction of housing, commerce and services. The objectives mentioned in the PDM for this UOPG are the recovery of the interior of the block, allowing its crossing.

STUDY AREA DEFINITION AND CHARACTERIZATION

Car accessibility played a major factor in the delimitation of the study area, as proposed. In order to accomplish this, the nearest main streets (and their respective directions) were considered, which are those carrying through traffic linking the various parts of the city. They were identified in a detailed plan of the area and this resulted in two rings (with the directions E, S, W, N, E and W, S, E, N, W, respectively). The major ring resulted in the actual delimitation of the study area, as it can seen above.

The main streets of the identified outer ring are:

- Constituição (*W*)
- Camões (*N and S*)
- Fernandes Tomás (*W*)
- Fernão Magalhães (*N and S*)

The achieved study area is located in two parishes, Santo Ildefonso and Bonfim.

3 ACCESSIBILITY ANALYSIS

The study area in the structural level is delimited by important axes of accessibility within the city, they are Rua Latino Coelho, Rua da Alegria, Rua de Santa Catarina and Rua de Gonçalo Cristóvão. In the accessibility analysis, it was important to identify these primary roads as the traffic is mainly oriented by them and, as a second point of study, understand how the secondary streets are articulated in order to penetrate the intervention area .

As the intervention area is a large block with high building density, the identification of these streets was an important factor when it comes to assess its permeability. The extension of the block is of 585 meters (N-S), which can be a conditioning factor for this aspect, posing some issues for the users.

According to the analysis (see Figure 3), four main streets were identified as accessibility axes, as their role is to distribute the traffic throughout the city. Then, the local distributors were also identified, which are the links that connect the immediate surroundings of the intervention area to the study area delimited previously.

In terms of evaluating the structural accessibility, as the study area is delimited by a considerable number of main streets, the ways to reach there by vehicle are well articulated, giving a good amount of choices for the users. Moreover, it is easily accessible by public transport provided by nearest metro stations, and also the bus interfaces, such as Trindade and Marquês (see Figure 3).

The streets worth mentioning, as it can be seen in Figure 3, are:

- Rua de Latino Coelho, with approximately 6m width;
- Rua de Santa Catarina, with approximately 7m width;
- Rua da Alegria, with approximately 8m width;
- Rua de Gonçalo Cristóvão, with 16m width.

The approximate width of a main street for an all purpose road is 7.6m, allowing the traffic flow of 1500 vehicles per hour. Thus, it can be assumed that these axes converging to the area bring a high level of daily traffic. Regarding the residential aspect of the study area, there should be an unsettled aspect, once the sidewalks are very narrow for the pedestrian walkability.

Focusing on the local accessibility, the block with high dimensions and closed nature brings a great concern when it comes to the quality of the built environment and its typology.

Combined with the level of density and the enclosure of private spaces inside the block, the quality of pedestrian ways to cross and navigate the public and private spaces in the block itself and the immediate surroundings is damaged.

On the southern part of the block it's noticeable that the density of buildings is not a relevant issue, but still there is a high level of enclosure between the inside and outside. This create a damaged aspects for the legibility and permeability. Regarding the uses, it has a main private use but the commercial activity doesn't relate with the public space, creating strong passive areas in the south part.

4 RELEVANT LANDMARKS

The intervention area at hand includes *Rua das Doze Casas* - former *Viela das Doze Casas* - whose name appeared in 1820, given that there would be this number of houses in the place at that time. In 1864, the alley was widened and aligned, and the *Centro Hípico do Porto* (Porto Equestrian Center) was also situated there. With a inconstant and misaligned profile, it has urban fronts with residential buildings and different services (such as workshops) of the second half of the XX century. In the surroundings of *Rua Latino Coelho*, the urban shapes are more stabilized, multifunctional, although with residential predominance.

While analysing the study and intervention area, based on relevant landmarks (Figure 4), it was pointed out that the immediate surrounding space contains four schools: *Colégio da Nossa Senhora da Paz* (number 8), *Externato Ribadouro* (number 12) and *ESMAE - Escola Superior de Música e Artes do Espectáculo* (number 1) and a public high school *Aurélia de Sousa* located outside the internal ring in a northeaster.

When it comes to public edification, we have *Administração Regional de Saúde do Norte* (number 11). Out of the intervention area, on the east, it is situated the water tower of *Monte dos Congregados* (number 5), which is still in use as a facility of the water network. And another building worth mentioning is *Direcção de Finanças*, which is located outside the intervention area, namely in the southwestern part (number 14).

In the private sector, there is a building which attracts people in this area, a nightclub called Eskada (number 4). It is located in the southeastern part of the intervention area, near *Rua da Alegria*. Two commercial centers located nearby, one inside the intervention area and the other in its immediate surroundings (number 9 and 10) .

The green private spaces in the insterstice of the considered block are badly maintained, especially in its southernmost part, which contain around 6000 m^2 of damaged yard. An important green area in the immediate surroundings is the garden José Moreira Silva which is located in the southeastern area close to the intervention area (number 2). To the north, the situation changes visibly, offering more greenery but summarizing this, the trees are less in number.

Though its size is not considerable, the intervention area accommodates one private parking lot with around 540 m^2 and two smaller ones with around 350 m^2 each of them.The core of the block at hand is mostly closed, bounded by walls in parts where buildings are not in place. It does not offer a physical nor visual connection between public and private areas. Paths

crossing the intervention area seem to be missing and legibility is in a very poor condition. And walking in the sidewalks of this area it is very perturb to know what exactly is happening beyond the walls of our city?

5 PRELIMINARY PROGRAM

DIAGNOSTIC ANALYSIS

Variety

A good urban space offers a variety of uses and experimental choices. It's possible to state that in the area of study, there are some parts with a good distribution of different uses and activities, most of them are part of *Rua de Santa Catarina*, what leads the place to have a good variety, influencing directly on the quality of the environment.

Although there is also an area, which is part of *Rua das Doze Casas*, with a few number of activities, where most of the buildings are residential ones. As a result, this part of the intervention area has a lower level of quality, as it's an urban space that is not able to attract people. It creates a problem, as only the ones who live there have a purpose to use this urban space, leading it to be a space with less capacity of choices, decreasing the quality of the environment.

Permeability

The permeability of an urban space depends highly on the number of alternative routes. These routes must be visible, making both physical and visual permeability important and thus, smaller blocks enable a better permeability, being 100 meters an ideal estimation of the larger dimension of a block.

The intervention area at hand consists of a block of large dimensions, as it was previously mentioned, with 585 meters of extension, when ideally it would be of around 100 meters. Moreover, this block presents dense and high facades (more than three stories high) along most of this perimeter with no streets nor paths crossing it, making it both physically and visually impermeable, with a poor private/public interface.

With this in mind, the green area identified during the characterization of the landmarks existent in the intervention are holds a relevant informal path which connects the western part with the eastern, emphasizing the lack of permeability felt by the users, which led them to create alternative routes on their own accord.

Legibility

The city should be well oriented, readable, easily walkable and a person should be highly aware of its environment. The concept of legibility of the cityscape is to create a more understandable area and it is never sufficient. The key physical elements which legibility contains are: nodes, edges, paths, districts and landmarks. Based on the analysis of our study area, all these elements are necessary and should be improved.

Among other things, visuality through eyes it is not enabled because of the boundaries created by residential buildings and walls. This leads us to conclude that inside the intervention area is not feasible to be record or notice any public activity, not even walking. While using the sidewalk nearby the wall of our intervention area, it is evident that the sense of getting somewhere is faded. Our area needs an image more vivid.

Robustness

The robustness in the city can be seen as the capacity to adapt through the changes, either in a social or economic dimension, that space can turn it function into a more sustainable use at any given moment. Once the robustness can be analysis method apply in different types of built environment, it can be divided into a large and small scale. Regarding the results of the analysis performed in the study area, the consent about the robustness lead to the diagnosis of the aspects to focus on.

For instance, a concerning point was the enclosure created by the private activities, creating zones incapable of contributing to the outdoors, meaning the street. Hence, the enclosure is reinforced by the width of the buildings and its density throughout the area of intervention. So in further discussion, the private gardens take major role in the articulation of problems as density combined with lack of active spaces and it uses.

OBJECTIVES

According to the diagnostic analysis of the area, the proposed objectives and respective actions to achieve them are the following:

- Improve legibility
    - Improve the private gardens
    - Improve the articulation between public and private areas
- Improve permeability
    - Creation of new axes of penetration in the block
    - Reduce the density in the diagnosed zone

Therefore, in order to achieve the objectives proposed above, the block was overlook with an attempt to see the best buildings, whether in location, whether in conservation state, to infiltrate the block. This penetration axes was thought to be a key in improving the permeability of the extent block street. And since the high density of the area, diagnosed as a relevant issue, it was delimited the spatialization of the question in order to have further solutions.

When improving the legibility, two main new streets was on the draft of important center lines on proposing some responses, and according to the diagnosis the area can have some potentialities in exploring new uses of the buildings. Some drawn attention was taken about the expropriation of the residential buildings, once the streets have schools and multi familiar entrances, the social aspects was taken into account in order to think in alternatives that could overcome this.

6 IMPLANTATION AREAS

The main challenge in drawing responses to the problems that area face, is to found infiltration areas that can be used as driven force for potentialities of the block. And among many solutions found during the studies, the first idea developed was a two main infiltration axes in the study are.

Thinking about the best perpendicular lines to open new streets, the analyses signalize for an new residential building, that could serve as a main extension to work on new uses, located in the north part of the block, near the gallery. The second one was the site near a nightclub building, that has a huge open air parking lot. In these sense, the only two streets could give a relative good extension to function new uses and activities.

An progress of the first developed idea, appropriates of the high density level of the Rua de Doze Casas, and the penetrative axes was the extension of the existent Street. The second penetration to open the block for a more legible aspects, was through the hotels. This axes show to be a important one, in order to add more value to the hotels, creating a new facade for possibility of improving the active areas. The third axes was the same in the first draft, showing the how strong is the mixed use potential for this area.

A lot was discussed about the potentialities of the private gardens, but having the important school in the area, became impracticable to think of ways of penetrating the block inside through north-south axes in the first drafts. But due to the problematic of density in Rua das Doze Casas, a response for that was expropriated those buildings, since most of them showed to be abandoned.

Since the improvement of the private/public relation was a main concern in drafting a proposal, the uses of the new buildings has a important role in it. In order to combine the residential aspects with no conflitant uses, the idea is to propose only buildings for commercial rooms, and the local commerce services to keep only the ground level. This was thought to improve the economic viability of the proposal as a whole. When the expropriation was a careful matter, the built environment proposed was taken care to add value whether in the social dimension and economic dimension.

7 ECONOMIC VIABILITY

The intervention has a positive revenue in the economic terms. It has an estimate cost of 14 millions euros, being of most of it (almost 5 millions) part of the fact that for this intervention it will be necessary to buy private lands, once the problems of the zone can only be solved with a physical intervention. The revenues of the buildings will be very positive, not only to improve functional aspects but also to get money enough to pay the costs. To draw this, it was made two tables in order to express the estimated values that the intervention will cost and how much it is possible to expect to profit from this in the nearest time, and then, finally, a balance of the profit and the renues.

Type	Cost			Average cost	Area	# of floors	Total Cost
	Standard	Medium	Superior	(€)	(m²)		(€)
Land							
Urbanized	€200,00		€400,00	€300,00	*16288*		€4.886.400,00
Not urbanized	€400,00		€800,00	€600,00			€0,00
Infrastructures							
Public space	€160,00			€160,00	*5955*		€952.800,00
Green space	€70,00			€70,00			€0,00
Buildings							
Urbanized land	€600,00	€750,00	€900,00	€750,00			€0,00
Multifamilar housing		€750,00	€100,00	€425,00	*3984*	*2*	€3.386.400,00
Unifamiliar housing	€300,00		€400,00	€350,00			€0,00
Comercial areas	€400,00		€500,00	€450,00	*7104*	*1*	€3.196.800,00
Offices	€250,00			€250,00		*1*	€0,00
Garages	€200,00			€200,00	*5544*	*1*	€1.108.800,00
Underground parking	€100,00			€100,00			€0,00
					42859		**€13.531.200,00**

Table 1 - Estimated costs

Type	Price		Average cost	Area	# of floors	Total Price
	Standard	Superior	(€)	(m²)		(€)
Multifamiliar	€700,00	€1.200,00	€950,00	3984	2	€7.569.600,00
Unifamiliar	€2.000,00	€6.000,00	€4.000,00	0	0	€0,00
Comercial areas	€3.000,00	€5.000,00	€4.000,00	7104	1	€28.416.000,00
Office areas	€1.500,00	€3.000,00	€2.250,00	0	1	€0,00
Garages	€1.500,00	€3.000,00	€2.250,00	5544	1	€12.474.000,00
Underground parking	€900,00	€1.200,00	€1.050,00	0	0	€0,00
Demolitions	€700,00	€900,00	€800,00			€0,00
				20616		**€48.459.600,00**

Table 2 - Estimated revenues

Balance	**€34.928.400,00**

Table 3 - Balance between costs and revenues

8 LEGIBILITY EVALUATION

As mentioned above the legibility of the neighborhood is never sufficient and the main aim of the proposals is increment of legibility on the study zone. Because of the density of residential buildings, physical elements of the legibility in some area, were completely missing thus, with that in mind, the first proposal was launched.

The whole zone is divided in four districts, creating a smaller blocks between communities. The boundaries between the first and second districts (north part of our study zone) are marked with a created path, which has it starts from the street R. Das Doze Casas and ends at the street R. de Santa Catarina. This new created path open the zone to be more readable and inviting for the surrounding area. It also increases the value of a landmark which is noted to be the Gallery. Furthermore, the new created buildings offer for community a small and acceptable blocks which include the privacy for people living there, and connection with outdoor area.

Part of the second district is located the school which in facade of the west part of the study area contains only one entry, all the activities are hidden inside the zone. Thus, to be able to create legibility out of it, the above mentioned proposed path, not only create small and readable blocks, but also it helps a lot into linking the northern part with center and the school would be able to breath.

Going down into the south part of the study zone, we can reach the next proposal which also as e main factor it contains increment of legibility. The new division between the second and third district are set by the new path, by which the creation of a node takes place in east part of the study area, and the edge closes this path in a west part. Meaning that this path is linking the two above mentioned street R. Das Doze Casas and R. de Santa Catarina with a high potential of highlighting the landmarks of two hotels. They are located near each other, one is belongs to Rua Das Doze Casas and the other is part of Rua de Santa Catarina. In the center zone of this created path, new buildings are created which in center they contain green area. This new block is part of the second district.

The creation of the buildings which are six in number, with the same structure and function near of each other, lead us with a new small and readable district on all sides. The green area is implemented in between which immediately gives a more vivid image of the neighborhood.

The proposal contains also the demolishment of the wall between the private and public sector. Not only that, but a new street in this southern linking Rua da Alegria and Rua de

Santa Catarina offers a more fluid walk and definitely more readable zone from outside too. This suggested part automatically connect the two sectors, private and public to be connected visually and physically. Since the length between two walls contains the distance around 40m in each side (west-east), greenery was suggested in a plan likewise, by offering a more inviting and a city-friendly area. Moreover this green area gives an opportunity for people living there to save and contain their privacy but also helping the environment and landscape of the city too.

9 ROBUSTNESS EVALUATION

Flexible buildings offer their users the opportunity to change and customise their homes, stores and workplaces. They can be altered to suit the users' individual preferences and changes in use. Much is down to the configuration of the building (its height, width and depth), access arrangements, the amount and configuration of internal space, and the way that thresholds are dealt with. As to the proposed solution, several aspects were taken into consideration, always with robustness in mind: building width, height and accesses.

BUILDING WIDTH

Building width has an impact on overall flexibility of use and the ability to personalise spaces, as well as on the vertical rhythms and amount of active frontage when seen in elevation.

According to the *Urban Design Compendium* made by the English Partnerships (National Regeneration Agency), buildings comprising 5 to 7 meters wide daylit "*cells*" or "*modules*" are able to provide a more flexible form. Each cell can be combined as desired and relates to a small shop or terraced house, for instance. Below these numbers, it will be more difficult to add rear extensions without blocking light and ventilation. The different proposed buildings follow these guidelines, as it is shown in Figure 8 and 9, enabling the diversity of uses and more varied streets.

BUILDING DEPTH

Building depth has an important impact on the need for artificial lighting and ventilation and this affects the diversity of uses that can be accommodated. Table 1, adapted from the aforementioned *Urban Design Compendium* provides a guide for assessing the impact of building depth on natural ventilation and lighting, and hence robustness.

Table 1 - The Implications of building depth (Source: Urban Design Compendium)

Building depth	Implications
< 9m	Too shallow for a central corridor and limited flexibility in internal planning
9-13 m	Provides naturally lit and ventilated space = OPTIMUM ROBUSTNESS
14-15 m	Sub-division is still facilitated, but some artificial ventilation and more artificial lighting is required
16-22m	More energy intensive, though a double-aspect cellular form is possible with the insertion of an atrium/light well, giving a block width of up to 40 m.

All buildings in this section were considered to have a depth of 12 m, approximately.

BUILDING HEIGHT

Tall buildings often have a positive influence when it comes to giving meaning to locations of important urban activity such as town centres, for instance. However, this has to be weighed against the possible negative impacts on the microclimate (such as wind funnels or too much shade in the streets), environmental performance of nearby buildings and amount of active facade, as it results in the risk of having fewer entrances and a more distanced relationship between users and the street itself.

Buildings with three or four stories in this context, provide an optimal solution, due to their ability to accommodate a range of uses (which generally decreases beyond four floors). They also have the potential for medium-high densities, and generally also enable lower energy demands and construction costs. For this proposal, all building were considered to have four floors with 2 different types of activities.

ACCESSES

The buildings must be sited to allow easy external access to as much as possible of the ground floor. The direct facade between buildings and public space should be maximised, to allow as many separate front doors as possible. In this project, the latter was followed in the building of District 1, but to minimize demolitions in Districts 2 and 3, the back facade is not accessible.

Separate entries from the street to upper levels were considered and its position should be place so as not to break up ground floor retail continuity.